PROJECT DOCUMENTATION

Scalability & Future Plan

Date	27 October 2026	**Team ID**	
Project Name	Global Fuel Prices and Taxation Trends: A Comprehensive Analysis	**Maximum Marks**	1 Mark

Scalability & Future Plan

This document identifies the current limits of the implemented fuel-price analysis project and proposes practical improvements. The current solution uses a CSV dataset, Tableau for analysis and visualization, Tableau Public for publishing, and a Flask-based website for project information. The items below are proposed future improvements, not features claimed to be already implemented.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	The analysis is based on the available CSV snapshot and its 2020–2026 coverage.	Newer records are not automatically included; findings depend on the dataset provided.	High
2	Tableau Public and the website depend on internet access for published content.	Users with limited connectivity may not be able to access dashboards and linked resources.	Medium
3	The analysis is descriptive and depends on the available definitions and fields.	Visual associations cannot prove that taxes, subsidies or crude oil alone caused price changes.	High

Scalability Plan

S.No	Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Dashboard and project information are shared through Tableau Public and a Flask website.	If usage grows, evaluate hosting the website on a reliable production platform and monitor traffic and errors.
2	Data Storage	The analysis uses a CSV file as its data source.	For larger or frequently updated datasets, assess a managed database or data warehouse; this is not part of the current implementation.
3	Performance	Performance depends on the dataset, workbook complexity, browser, and network.	Reduce unnecessary marks and calculations, review filters, optimize workbook design, and record actual load times before and after changes.
4	Security	Published Tableau content is publicly accessible according to its sharing settings.	Avoid publishing sensitive data; review sharing permissions and apply authentication to any future private website or data service.

Future Roadmap

The timeline below is a suggested roadmap after the current project submission. It is not a commitment that these enhancements have already been started.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Refresh and document the source dataset; repeat quality checks and update Tableau views when new data becomes available.	1–2 months after submission	More current and consistent analysis.
Phase 2	Add deeper comparisons by region, income level, tax and subsidy categories, with clear definitions and caveats.	2–3 months	More useful comparisons for research and policy exploration.
Phase 3	Evaluate automated data ingestion and a scheduled refresh workflow if a trustworthy update source is available.	3–6 months	Reduced manual updates and improved repeatability.
Phase 4	Conduct structured usability and performance tests; consider access controls or a larger data platform only if project needs justify them.	6 months or later	Improved reliability, usability and readiness for larger use.

Recommended Success Measures

• Dataset refresh: document source, retrieval date, covered period, and any schema changes.

• Data quality: rerun missing-value, duplicate, field-type, and range checks after each refresh.

• Dashboard performance: record measured initial load and filter-response times under a repeatable test setup; do not report estimates as measured results.

• Usability: gather feedback from representative users and record any confusing labels or navigation issues.

• Reliability: check that published Tableau links and the project website remain accessible.

Conclusion

The current project is suitable as a descriptive analytics prototype built around a CSV dataset, Tableau dashboard and Story, and a supporting Flask website. The most practical next steps are to document data provenance, refresh data when a reliable source is available, validate quality after updates, and measure performance before making larger infrastructure changes. A cloud warehouse or automated pipeline should be considered only if future data volume or refresh requirements justify it.

Team ID is intentionally blank. Update the roadmap after discussing priorities with the project team.